Calculus & Statistics

AIGIS Study Pack - Full Edition

Built by AIGIS (Accurately Imparting Information Sensei) for Naufal Arham Maliky.
19 units covering the full university curriculum - sequences, series, vectors, linear algebra, multiple integrals, joint distributions, ANOVA and more.
Geodesy is ONE application among physics, engineering, CS and everyday use.

All formulas are typeset with real mathematical notation (fractions, limits, integrals, Greek letters) - not computer ASCII. Companion Obsidian notes (Mathematics/) sync to Google Drive for NotebookLM.

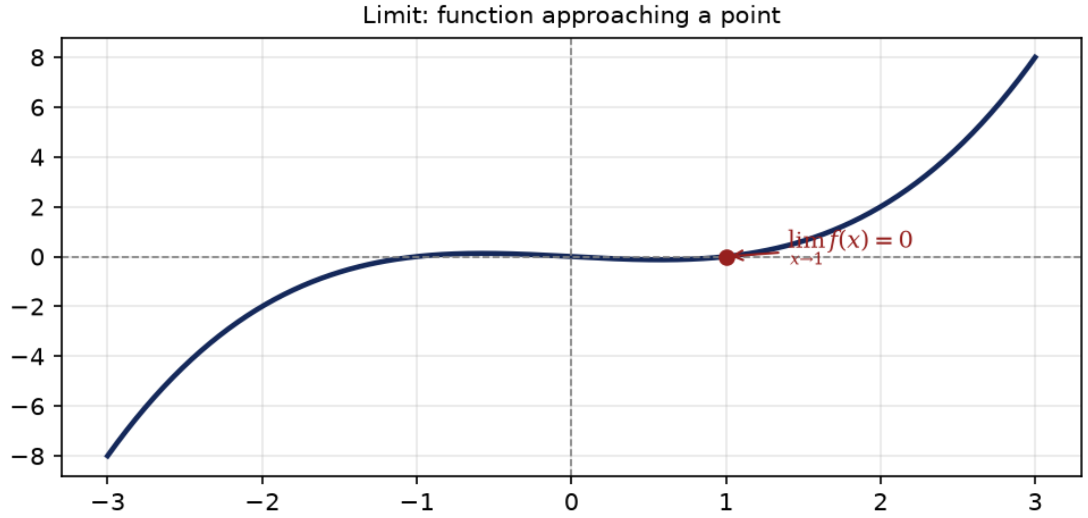
Sources: Stewart (Calculus), Walpole (Probability & Statistics), Lay (Linear Algebra), Vermeer (Geodesy).

Contents

Includes Glossary and Geodesy / Physics / Engineering / CS application index.

1. Limits & Continuity

The limit is the value f(x) approaches as x approaches a. Continuity (no jumps, gaps or asymptotes) is required before we can differentiate.



Key formulas

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$$

$$\lim_{x\to a} f(x) = L$$

$$\text{continuous at } a: \lim_{x\to a} f(x) = f(a)$$

$$\text{L'Hopital: } \lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)} \quad (0/0,\ \infty/\infty)$$

Worked example

Example:

$$Find \lim_{x\to 0} \frac{\sin x}{x}$$

Step 1: Substitute x=0 gives 0/0, so use L'Hopital.

Step 2: Derivative top: cos x; bottom: 1.

Step 3:

$$\lim_{x\to 0} \frac{\cos x}{1} = \cos 0 = 1$$

Where it is used

Foundation of numerical differentiation and iteration convergence.

Applications: Geodesy: least-squares iteration | Physics: rates of change | Engineering: error analysis

-> Vault note: [[Least Squares Adjustment]]

2. Derivatives

The derivative is the instantaneous rate of change - the slope of the tangent line. It underpins optimisation, sensitivity and linearisation.

Key formulas

$$\frac{d}{dx}x^n = nx^{n-1}$$

$$\frac{d}{dx}\sin x = \cos x, \quad \frac{d}{dx}e^x = e^x$$

$$\frac{d}{dx}\ln x = \frac{1}{x}$$

$$\text{Chain: } (f \circ g)' = f'(g)\,g'$$

$$\text{Product: } (uv)' = u'v + uv'$$

$$\frac{\partial f}{\partial x} \text{ holds other variables constant}$$

Worked example

Example:

$$Differentiate f(x) = x^2 \sin x$$

Step 1:

$$Product rule : u = x^2, v = sinx$$

Step 2: u'=2x, v'=cos x.

Step 3:

$$f' = 2x\sin x + x^2\cos x$$

Where it is used

Linearises models and builds Jacobians of observation equations.

Applications: Geodesy: Newton-Raphson, Jacobian A | Physics: velocity = dx/dt | Economics: marginal cost
-> Vault note: [[Least Squares Adjustment]]

3. Applications of Derivatives

Derivatives locate maxima, minima and tangents, and drive Newton-Raphson root finding.

Key formulas

$$\text{critical point: } f'(x) = 0$$

$$\text{min if } f''(x) > 0, \quad \text{max if } f''(x) < 0$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$f(x+h) = f + hf' + \frac{h^2}{2}f'' + \cdots \text{ (Taylor)}$$

Worked example

Example:

$$Minimise f(x) = x^2 - 4x + 5$$

Step 1:

$$f'(x) = 2x - 4 = 0 \Rightarrow x = 2$$

Step 2:

$$f''(x) = 2 > 0 \Rightarrow \text{minimum}$$

Step 3:

$$f(2) = 1. \ \text{Min at } (2, 1)$$

Where it is used

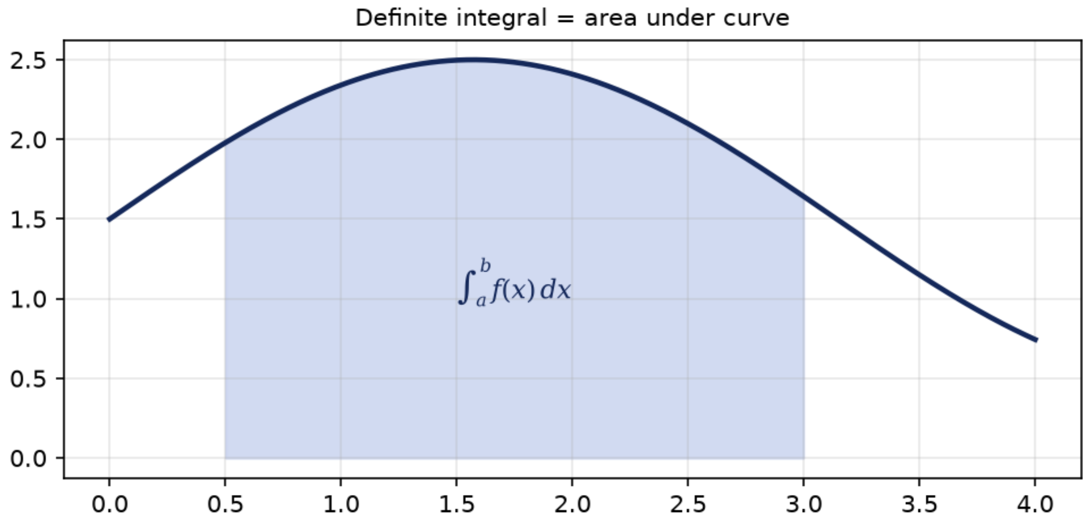
Solving unknown parameters; GNSS ambiguity fixing.

Applications: Geodesy: GPS ambiguity | Physics: equilibrium points | ML: gradient descent

-> Vault note: [[Geoid Undulation]]

4. Integrals

The integral is the antiderivative and the signed area under a curve. The Fundamental Theorem links the two.



Key formulas

$$\int f'(x)\,dx = f(x) + C$$

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

$$\int f(g(x))\,g'(x)\,dx = \int f(u)\,du$$

$$\int u\,dv = uv - \int v\,du \text{ (by parts)}$$

Worked example

Example:

$$Evaluate \int_0^2 2x\,dx$$

Step 1:

$$Antiderivative : x^2$$

Step 2: F(2)-F(0)=4-0=4.

Check: Triangle base 2, height 4: area = 4. Matches.

Where it is used

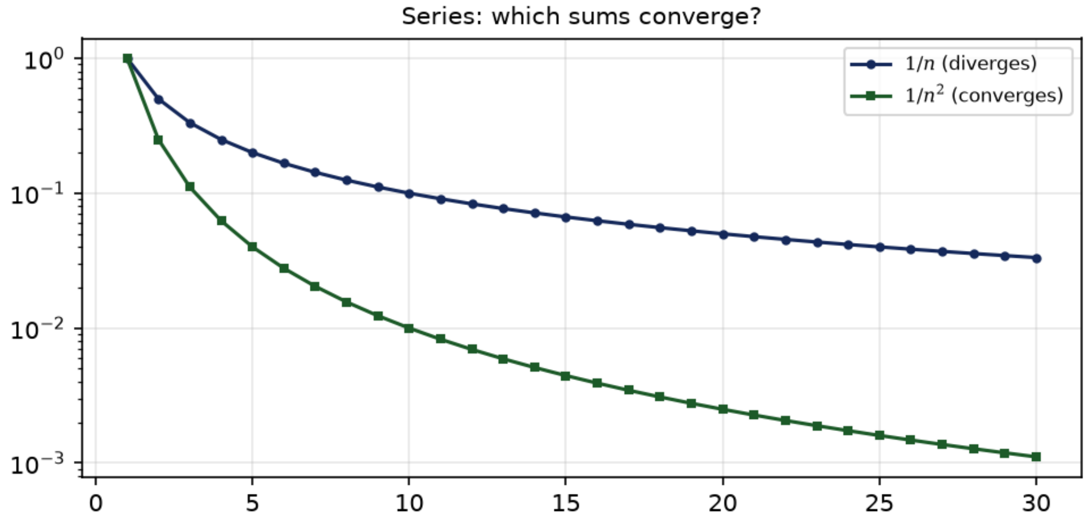
Volumes/areas in terrain models; probability from density.

Applications: Geodesy: DTM volumes | Physics: work = integral of force | Probability: CDF

-> Vault note: [[Digital Terrain Models|Geodesy ch.9]]

5. Sequences & Series

A sequence is an ordered list; a series is its sum. Convergence decides whether the sum is finite.



Key formulas

$$\text{Sequence: } a_1, a_2, \ldots, a_n$$

$$\text{Series: } S = \sum_{n=1}^{\infty} a_n$$

$$\text{Geometric: } \sum_{n=0}^{\infty} ar^n = \tfrac{a}{1-r} \; (|r| < 1)$$

$$\text{p-series: } \sum \tfrac{1}{n^p} \text{ converges if } p > 1$$

$$\text{Ratio test: } \lim \left|\tfrac{a_{n+1}}{a_n}\right| < 1 \Rightarrow \text{converges}$$

Worked example

Example:

$$Does \sum_{n=1}^{\infty} \tfrac{1}{n^2} converge?$$

Step 1: p-series with p=2.

Step 2: p>1, so it converges.

Note:

$$Sum = \pi^2/6 \approx 1.645$$

Where it is used

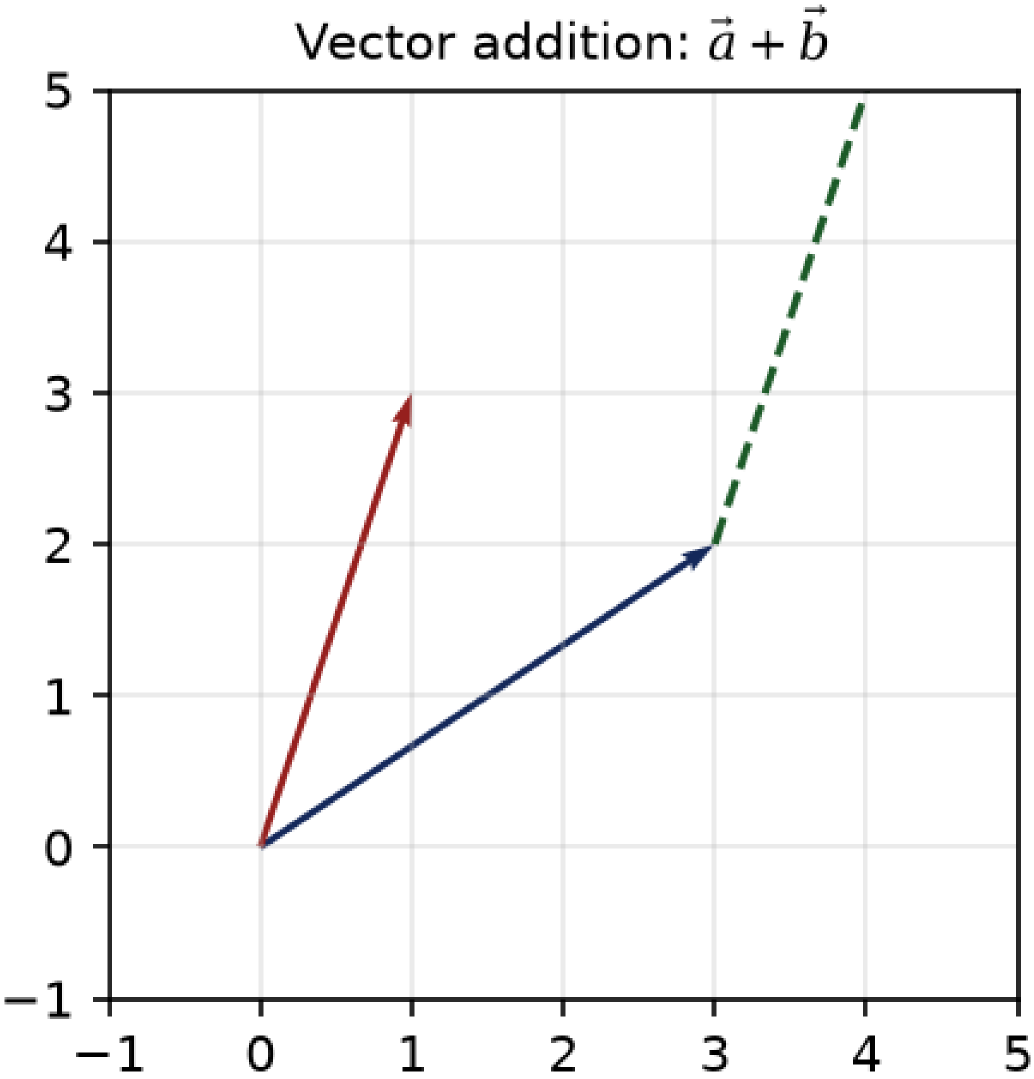
Fourier series; signal approximation; numerical methods.

Applications: Geodesy: harmonic expansions | Signal processing: FFT | CS: algorithm analysis

-> Vault note: [[Least Squares Adjustment]]

6. Vectors & Geometry

Vectors carry magnitude and direction - the language of forces, velocities and 3D coordinates.



Key formulas

$$\vec{a} = (a_1, a_2, a_3)$$

$$\vec{a}+\vec{b} = (a_1+b_1,\ a_2+b_2,\ a_3+b_3)$$

$$\vec{a}\cdot\vec{b} = |\vec{a}||\vec{b}|\cos\theta = \sum a_i b_i$$

$$|\vec{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$$

$$\vec{a} \times \vec{b} \text{ (perpendicular, } |\vec{a}||\vec{b}|\sin\theta)$$

Worked example

Example:

$$Anglebetween\vec{a} = (1,0,0)and\vec{b} = (1,1,0)?$$

Step 1:

$$\vec{a}\cdot\vec{b} = 1,\ |\vec{a}| = 1,\ |\vec{b}| = \sqrt{2}$$

Step 2:

$$\cos\theta = 1/\sqrt{2}$$

Step 3:

$$\theta = 45^\circ$$

Where it is used

3D coordinate frames; baseline vectors in GNSS.

Applications: Geodesy: ECEF vectors | Physics: force & velocity | Graphics: normals

-> Vault note: [[Geocentric Cartesian ECEF]]

7. Linear Algebra Foundations

Matrices organise systems of equations - central to least squares and transformations.

Matrix, determinant & inverse

$$A = [\,a \;\; b\,;\;\; c \;\; d\,]$$

$$\det A = ad - bc$$

$$A^{-1} = \frac{1}{ad - bc}[\,d \;\; -b\,;\;\; -c \;\; a\,]$$

Key formulas

$$A = [\,a \;\; b\,;\;\; c \;\; d\,] \quad (\text{rows } [a\ b], [c\ d])$$

$$\det A = ad - bc$$

$$A^{-1} = \frac{1}{ad - bc}[\,d \;\; -b\,;\;\; -c \;\; a\,]$$

$$A\vec{x} = \vec{b} \;\Rightarrow\; \vec{x} = A^{-1}\vec{b}$$

$$\text{Eigen: } A\vec{v} = \lambda\vec{v}$$

$$\text{Normal eq: } A^T P A\,\vec{x} = A^T P\vec{l}$$

Worked example

Example:

$$Solve 2x + y = 3,\ x + y = 2$$

Step 1:

$$\det = 2(1) - 1(1) = 1$$

Step 2:

$$x = 3 - 2 = 1,\ y = 2 - 1 = 1$$

Check:

$$2(1) + 1 = 3,\ 1 + 1 = 2$$

Where it is used

The normal equations of adjustment; datum transformations.
Applications: Geodesy: Helmert transform | CS: graphics & ML | Economics: input-output
-> Vault note: [[Helmert Transformation]]

8. Multivariable Calculus

Functions of several variables; gradients and Jacobians appear throughout science.

Key formulas

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right)$$

$$D_{\vec{u}} f = \nabla f \cdot \vec{u}$$

$$J = \left[\frac{\partial f_i}{\partial x_j}\right] \text{ (Jacobian)}$$

$$\frac{\partial^2 f}{\partial x_i \partial x_j}$$

Worked example

Example:

$$f(x, y) = x^2 + y^2. Gradientat(1, 1)?$$

Step 1:

$$\partial f/\partial x = 2x,\ \ \partial f/\partial y = 2y$$

Step 2:

$$\nabla f(1, 1) = (2, 2).\ \text{ Points uphill.}$$

Where it is used

Design matrix A; error propagation; map projections.

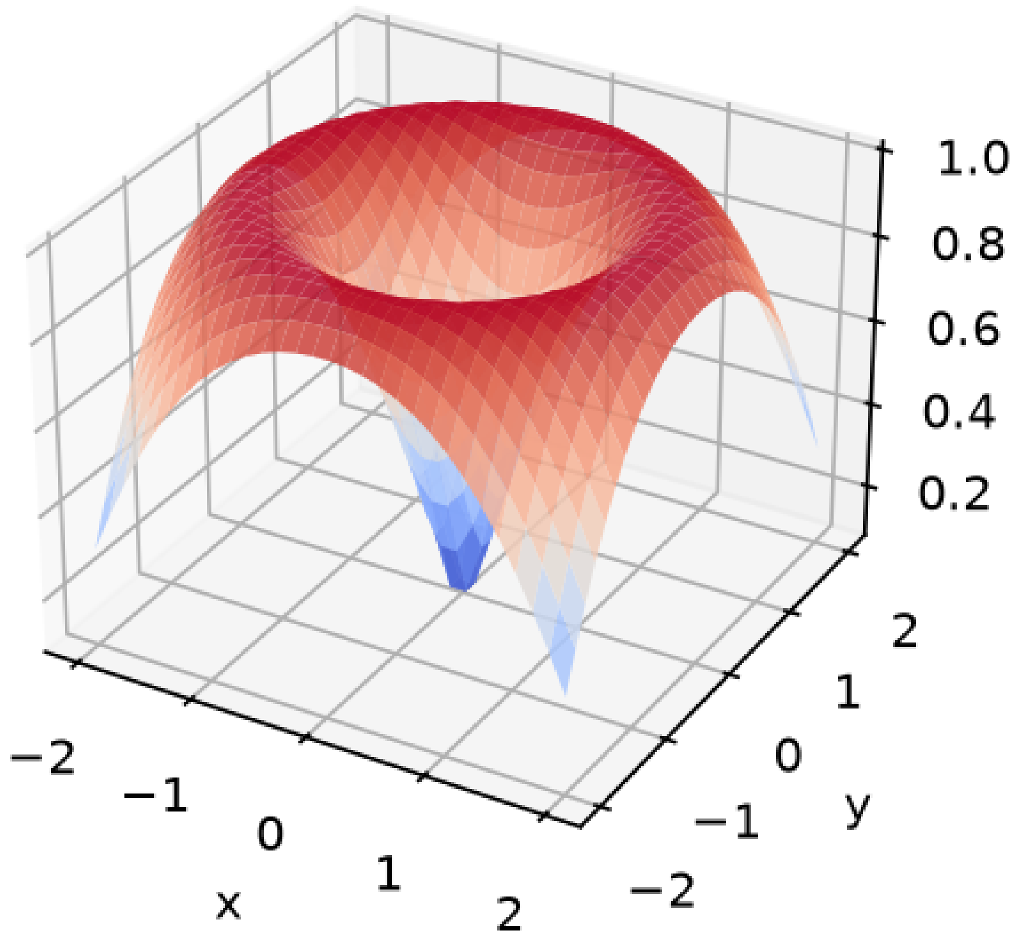
Applications: Geodesy: design matrix A | Physics: gradient fields | ML: backprop

-> Vault note: [[Helmert Transformation]]

9. Multiple Integrals

Integrate over areas and volumes; the workhorse of mass, probability and flux.

$$\iint_R f(x, y)\, dA$$



Key formulas

$$\iint_R f(x, y)\, dA = \int_a^b \int_{g(x)}^{h(x)} f(x, y)\, dy\, dx$$

$$\iiint_V f\, dV$$

$$\text{Polar: } dA = r\, dr\, d\theta$$

Change of vars: $dA = |J|\,du\,dv$

Worked example

Example: Area of unit disk via polar coords.

Step 1:

$$\iint r\,dr\,d\theta,\ r \in [0,1],\ \theta \in [0,2\pi]$$

Step 2:

$$\int_0^{2\pi}\int_0^1 r\,dr\,d\theta = \pi$$

Where it is used

Mass/volume; 2D probability; moments of inertia.

Applications: Geodesy: DTM volume | Physics: centre of mass | Probability: 2D CDF

-> Vault note: [[Digital Terrain Models|Geodesy ch.9]]

10. Differential Equations

Relations between a function and its derivatives; model change over time.

Key formulas

$$\frac{dy}{dx} = f(x) \text{ (separable)}$$

$$\frac{dy}{dx} + P(x)y = Q(x) \text{ (linear 1st order)}$$

$$\mu(x) = e^{\int P(x)\,dx} \text{ (integrating factor)}$$

$$\frac{d^2y}{dx^2} + k^2y = 0 \Rightarrow y = A\cos kx + B\sin kx$$

Worked example

Example:

$$Solve dy/dx = 2y,\ y(0) = 1$$

Step 1: Separate: dy/y = 2 dx.
Step 2: Integrate: ln|y| = 2x + C.
Step 3:

$$y = e^{2x} \text{ } (y(0) = 1 \Rightarrow C = 0)$$

Where it is used

Orbit dynamics; population; circuits; heat flow.

Applications: Geodesy: satellite orbits | Physics: springs & circuits | Biology: growth

-> Vault note: [[GPS]]

11. Descriptive Statistics

Summarise data by centre, spread and shape before inference.

Key formulas

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n} (x_i - \bar{x})^2$$

$$s = \sqrt{s^2}$$

$$\mathrm{Cov}(X, Y) = \frac{1}{n-1}\sum(x_i - \bar{x})(y_i - \bar{y})$$

$$r = \frac{\mathrm{Cov}(X, Y)}{s_x s_y}, \quad -1 \le r \le 1$$

Worked example

Example: Data: 2,4,4,4,5,5,7,9. Mean & std?

Step 1:

$$\bar{x} = 40/8 = 5$$

Step 2:

$$s^2 = \frac{32}{7} \approx 4.57$$

Step 3:

$$s \approx 2.14$$

Where it is used

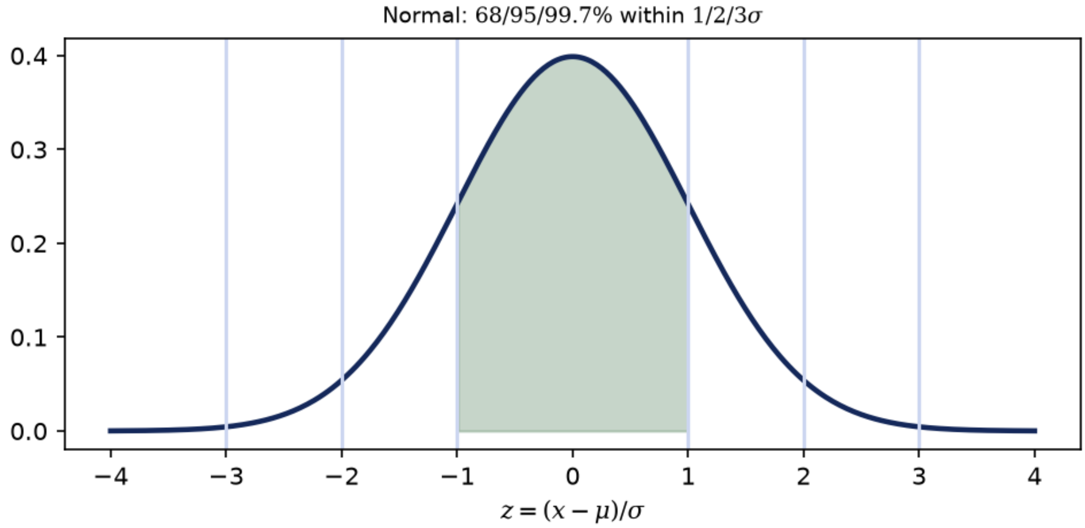
Analysing residuals; network reliability.

Applications: Geodesy: residuals | Every field: EDA | Business: KPIs

-> Vault note: [[Least Squares Adjustment]]

12. Probability Distributions

Models of random variation; the normal distribution dominates measurement error.



Key formulas

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$Z = \frac{X-\mu}{\sigma} \sim N(0, 1)$$

$$\approx 68/95/99.7\% \text{ within } 1/2/3\sigma$$

chi^2=sum_i=1^k Z_i^2,quad t=fracbar X-mus/sqrt n

textCLT: bar XsimNleft(mu,fracsigma^2nright)

Worked example

Example:

$$X \sim N(0, 1). FindP(-1 < X < 1)$$

Step 1: By the 68-95-99.7 rule, 1 sigma = 68%.
Answer: Approximately 0.68.

Where it is used

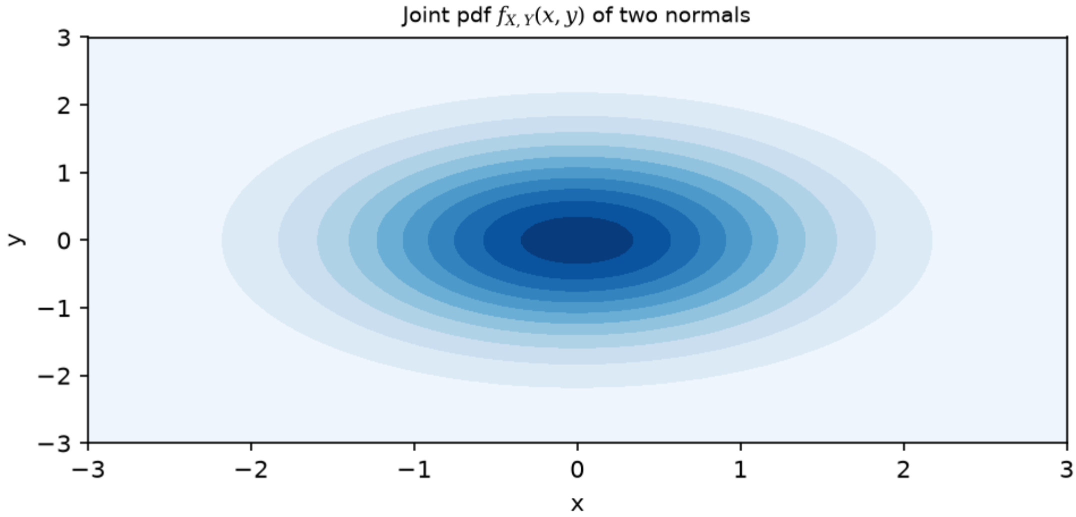
Error ellipse; statistical testing of adjustments.

Applications: Geodesy: error ellipse | Science: measurement | Finance: returns

-> Vault note: [[Least Squares Adjustment]]

13. Joint & Marginal Distributions

Two (or more) random variables together; marginals summarise one at a time.



Key formulas

$$f_{X,Y}(x,y) \geq 0, \quad \iint f_{X,Y} = 1$$

$$f_X(x) = \int f_{X,Y}(x,y)\,dy \text{ (marginal)}$$

$$\text{Independent if } f_{X,Y} = f_X f_Y$$

$$\mathrm{Cov}(X,Y) = E[(X-\mu_X)(Y-\mu_Y)]$$

$$\mathrm{Corr}(X,Y) = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y}$$

Worked example

Example: X,Y independent N(0,1). Cov(X,Y)?

Step 1: Independence => Cov = 0.

Step 2: Correlation = 0 too.

Note: Zero correlation != independence (in general).

Where it is used

Uncertainty of coordinate pairs; error ellipses.

Applications: Geodesy: error ellipse | ML: multivariate data | Genetics: linkage

-> Vault note: [[Least Squares Adjustment]]

14. Sampling & Estimation

Infer population properties from a sample.

Key formulas

$$\text{Bias} = E[\hat{\theta}] - \theta$$

textCI: bar x pm t_alpha/2,s/sqrt n

operatornameSE(bar x) = s/sqrt n

$$\text{MLE: maximise } L(\theta) = \prod f(x_i|\theta)$$

Worked example

Example:

$$n = 25,\ \bar{x} = 10,\ s = 2.\ \text{95\% CI for mean?}$$

Step 1:

$$t_{0.025,\,24} = 2.064,\ SE = 2/5 = 0.4$$

Step 2:

$$10 \pm 2.064(0.4) = 10 \pm 0.83$$

Where it is used

Precision of estimated coordinates; survey design.

Applications: Geodesy: network design | Polling: margins | QC: acceptance

-> Vault note: [[Datum Transformation]]

15. Hypothesis Testing

Decide whether data contradicts a null hypothesis H0.

Key formulas

$$H_0 \text{ vs } H_1; \quad p\text{-value}$$

$$\alpha = P(\text{Type I}), \quad \beta = P(\text{Type II})$$

$$\text{Reject } H_0 \text{ if } p < \alpha$$

t=fracbar x-mu_0s/sqrt n,quad chi^2=sumfrac(O-E)^2E

Worked example

Example: Test mean=0, n=10, t=2.5.

Step 1:

$$t_{0.025,\,9} = 2.262$$

Step 2: 2.5 > 2.262 => reject H0.

Note: p < 0.05; significant.

Where it is used

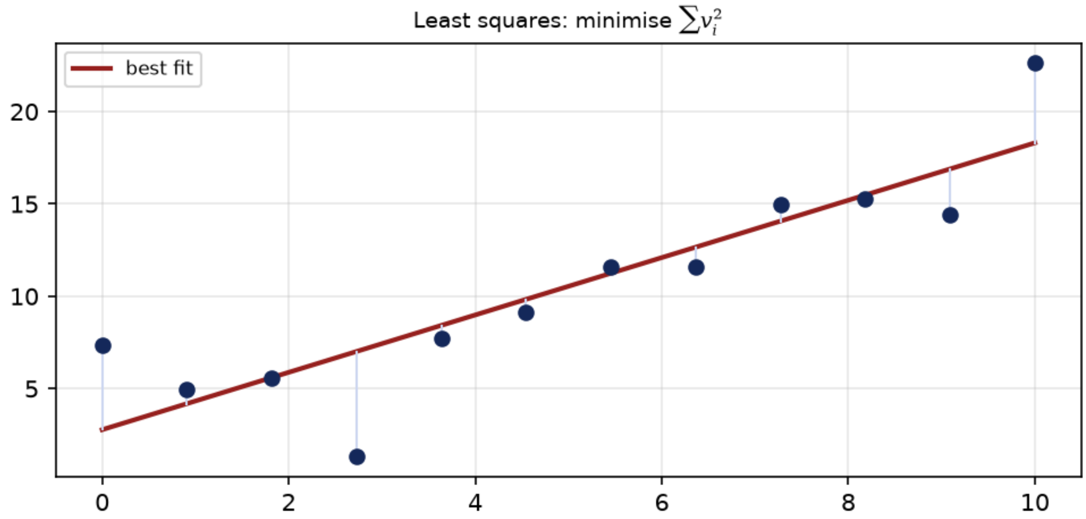
Global model validation; outlier detection.

Applications: Geodesy: Vermeer ch.14 | Medicine: trials | A/B testing

-> Vault note: [[Least Squares Adjustment]]

16. Regression & Least Squares

Fit a model to data by minimising weighted squared residuals - the core of geodesy.



Key formulas

$$\vec{l} = A\vec{x} + \vec{v}$$

$$\text{minimise } \vec{v}^T P\vec{v} \;\Rightarrow\; A^T PA\,\vec{x} = A^T P\vec{l}$$

$$\hat{\vec{x}} = (A^T PA)^{-1} A^T P\vec{l}$$

$$C_{\hat{x}} = \sigma_0^2 (A^T PA)^{-1}$$

$$\sigma_0^2 = \frac{\vec{v}^T P\vec{v}}{n-u}$$

Worked example

Example: Fit y=ax through origin to (1,2),(2,4),(3,5).

Step 1:

$$A = [1, 2, 3]^T,\; \vec{l} = [2, 4, 5]^T,\; P = I$$

Step 2:

$$a = \frac{A^T\vec{l}}{A^TPA} = \frac{25}{14}$$

Step 3:

$$a \approx 1.79$$

Where it is used

Every adjustment: GPS, levelling, traverse, datum transform.

Applications: Geodesy: ALL adjustments | Science: curve fit | Finance: trends

-> Vault note: [[Least Squares Adjustment]]

17. Error Propagation

How input uncertainty flows to outputs (law of covariance propagation).

Key formulas

$$y = f(\vec{x}), \quad \sigma_y^2 \approx \left(\frac{\partial f}{\partial \vec{x}}\right)^T C_x \left(\frac{\partial f}{\partial \vec{x}}\right)$$

$$y = ax + b \;\Rightarrow\; \sigma_y = |a|\sigma_x$$

$$\sigma_{x+y}^2 = \sigma_x^2 + \sigma_y^2 \text{ (independent)}$$

$$\left(\frac{\sigma_{xy}}{xy}\right)^2 = \left(\frac{\sigma_x}{x}\right)^2 + \left(\frac{\sigma_y}{y}\right)^2$$

Worked example

Example:

$$y = 3x,\ \sigma_x = 0.1. Find \sigma_y$$

Step 1:

$$Linear: \sigma_y = |3|\sigma_x$$

Step 2:

$$\sigma_y = 3(0.1) = 0.3$$

Where it is used

Coordinate precision; GNSS dilution of precision (DOP).

Applications: Geodesy: DOP | Engineering: tolerances | Metrology

-> Vault note: [[GNSS]]

18. ANOVA

Compare means across three or more groups in one test.

Key formulas

$$\text{Between: } SS_B = \sum n_j(\bar{x}_j - \bar{x})^2$$

$$\text{Within: } SS_W = \sum\sum(x_{ij} - \bar{x}_j)^2$$

$$F = \frac{SS_B/(k-1)}{SS_W/(N-k)}$$

$$\text{Reject } H_0 \text{ if } F > F_\alpha$$

Worked example

Example: Three survey methods, means 2,3,4.5. Equal?

Step 1:

$$Compute SS_B and SS_W$$

Step 2:

$$F = MS_B/MS_W$$

Step 3: Compare to F-critical; reject if large.

Where it is used

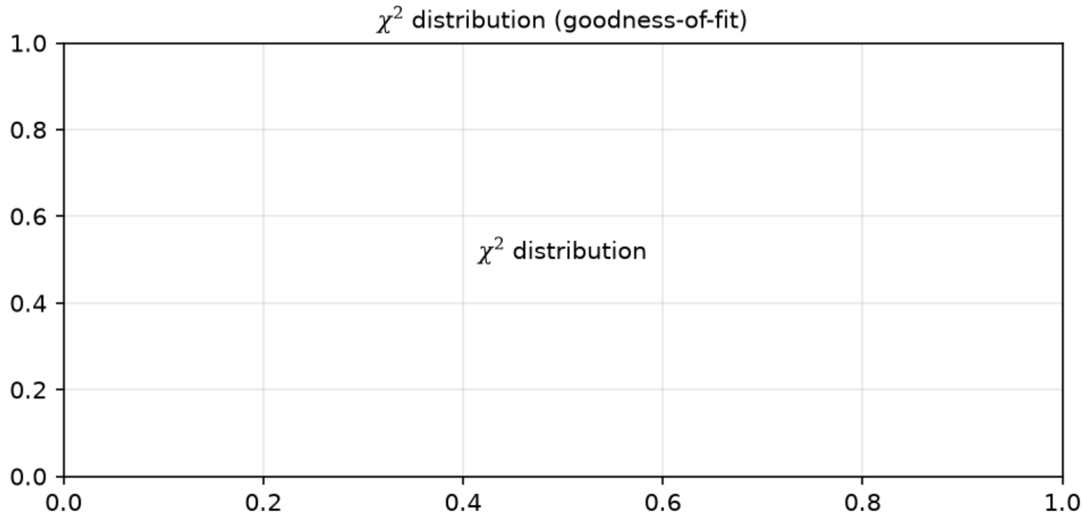
Comparing measurement techniques; sensor calibration.

Applications: Geodesy: instrument comparison | Agronomy: yields | Medicine: treatments

-> Vault note: [[Least Squares Adjustment]]

19. Chi-square & Non-parametric

Tests that don't assume normality: goodness-of-fit and rank-based methods.



Key formulas

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

$$\mathrm{df} = (\mathrm{rows} - 1)(\mathrm{cols} - 1)$$

$$\text{Mann-Whitney } U, \quad \text{Wilcoxon signed-rank}$$

$$\text{Spearman } \rho \text{ (rank correlation)}$$

Worked example

Example: Observed [30,20], expected [25,25]. Fit?

Step 1:

$$\chi^2 = (30 - 25)^2/25 + (20 - 25)^2/25 = 2$$

Step 2:

$$df = 1, \ \chi^2_{0.05} = 3.84$$

Step 3: 2 < 3.84 => not significant.

Where it is used

Goodness-of-fit for classified observations; robust stats.

Applications: Geodesy: classification tests | Ecology: counts | CS: A/B ranking

-> Vault note: [[Least Squares Adjustment]]

Glossary

Derivative	rate of change; slope of tangent
Integral	antiderivative; signed area
Gradient	vector of partial derivatives; points uphill
Jacobian	matrix of first partial derivatives
Residual	observed minus computed (v)
Variance	average squared deviation from mean
Covariance	how two variables co-vary
Confidence	interval likely containing the true value
p-value	probability of the result if H0 were true
Least squares	method minimising sum of squared residuals
Eigenvalue	scalar lambda with A v = lambda v
ANOVA	test comparing 3+ group means